



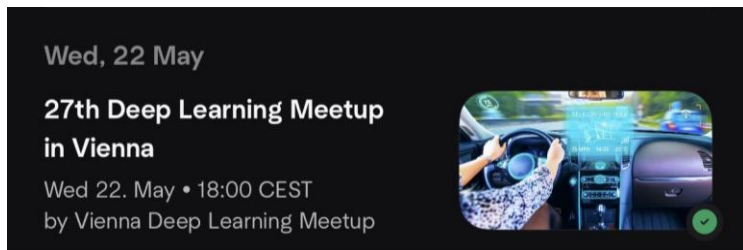
Cleared for Takeoff: Automatic Speech Recognition in Air Traffic Management

From research to product.

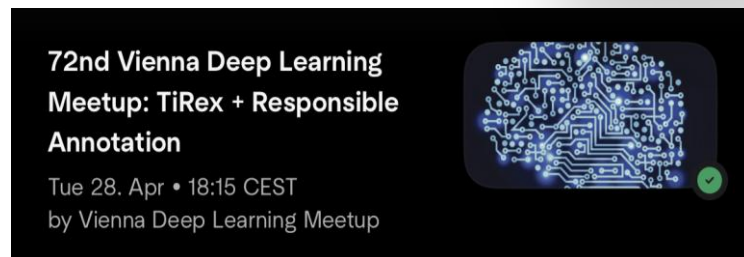
About Sahebeh (Saba):

Sahebeh (Saba) Dadboud – Expert Data Scientist @ Frequentis

- 8+ years in AI / Data Science
- Driving AI initiatives in AI Competence Centre
- Building **production-grade AI systems for safety-critical domains**
- Focus: **MLOps, GenAI, scalable AI systems**



2019



2026

About Christian:



Christian Soltzick – Audio Expert @ Frequentis

- Lead of HET **Audio Competence Centre**
- Improving audio quality across all Frequentis voice communication products
- Supporting product development related to automatic speech recognition and contributing to research projects, such as the FFG “Next Generation Safety” project with focus on AI-based automation in Air Traffic Management
- 15+ years of professional experience with background in the field of acoustics, including electroacoustic transducers, acoustic measurements, and signal processing

We set standards
in control centres worldwide

FREQUENTIS
FOR A SAFER WORLD

Enabling our global
customers to make the
world a safer place



... for more than 75 years!

2,600 employees

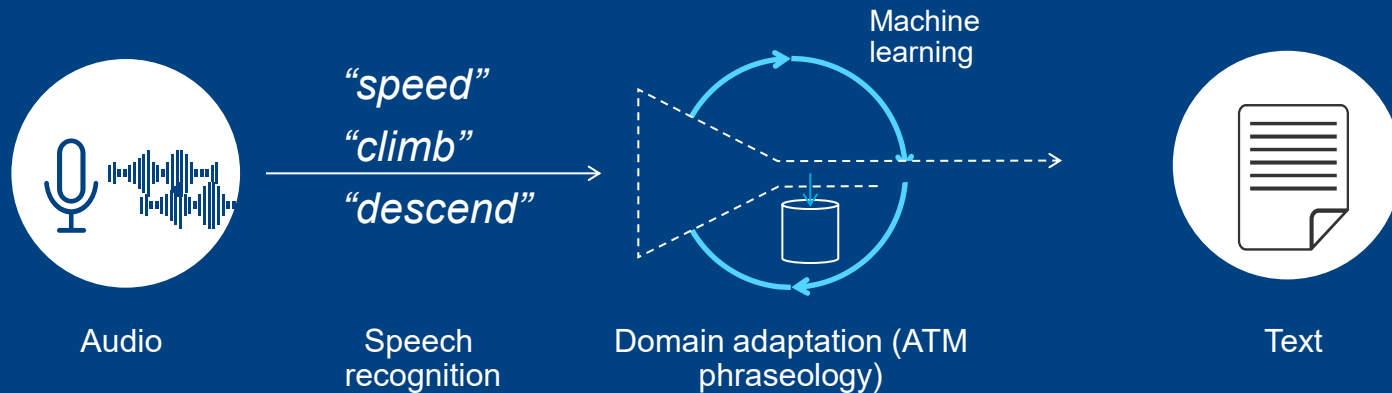
€580m revenues (preliminary)

Vienna headquarters

Our vision
#1 in safety-
critical
control centre
solutions



How automatic transcription should work!



First step to increased automation levels

Improve efficiency

Speeding up workflow for safety & quality

Machine learning

Boosts the accuracy of transcription constantly

Increase traffic handling capacity

Free ATCOs for critical tasks

Elevating ATCO to supervise output

Avoiding tedious/ routine work, improving focus & increasing quality

Automatic Speech Recognition – ATM Use Cases

Post processing

Safety investigation

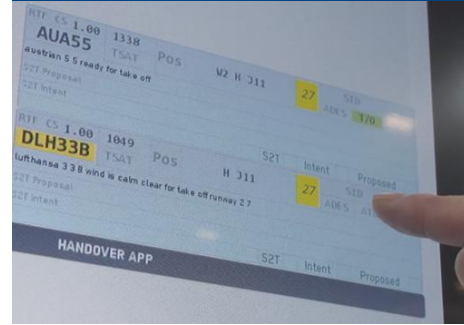
09:30 09:35:55 09:36:00 09:36:05 09:36:10 09:36:15 09:36:20 09:36:25 09:36:30 09:36:35 09:36:40

Hudson River Landing

09:32:08	LGA Dep	north west three thirty seven new york departure radar contact maintain one five thousand
09:32:13	NW 337	one five thousand north west three thirty seven
09:32:53	External	hello departure global three seven six golf four point four for seven thousand
09:32:59	LGA Dep	government seven six golf new york departure climb and maintain one five thousand
09:33:02	External	right up to one five thousand seven six golf
09:33:20	LGA Dep	north west three thirty seven turn left three six zero
09:33:24	LGA Dep	left three six zero north west three thirty seven
09:34:09	LGA Dep	government seven six golf fly heading of two six zero contact departure one two zero eight

Real time processing

Callsign highlighting



Readback error detection



KLM one zero four, climb flight level three six zero

Compare ATCO and pilot

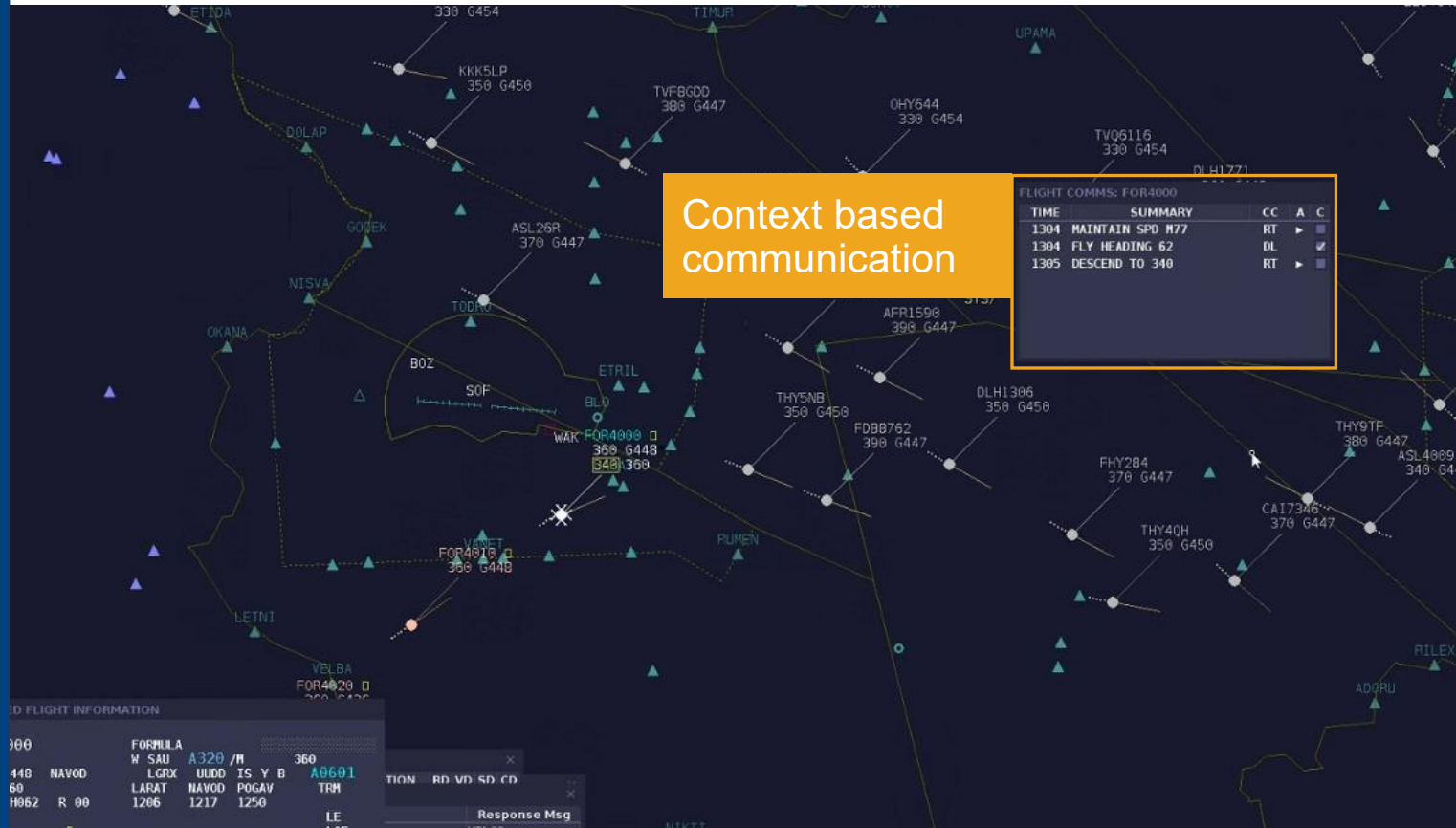


climb flight level three eight zero, KLM one zero four



Use Case

Integration into ATM system



Why ASR fails in reality?

Did you know: It takes about 10 hours to manually transcribe a 1-hour voice recording.



Word error rate too high

- COTS S2T engines not suitable → word error rate (WER) is between 25-50% for ATM communication
- Challenges with typical speech quality (VHF radio) and training data (reference transcription)
- S2T models in Frequentis GuardX are currently trained with +4000h
- Current GuardX base model has WER between 10-20%

Every customer environment is different

- Every customer / region is different (accents, dialects, audio quality, specific city names...)
- Training data does not contain all possible Callsigns or Waypoints
- It's difficult to get customer data (privacy / data security)

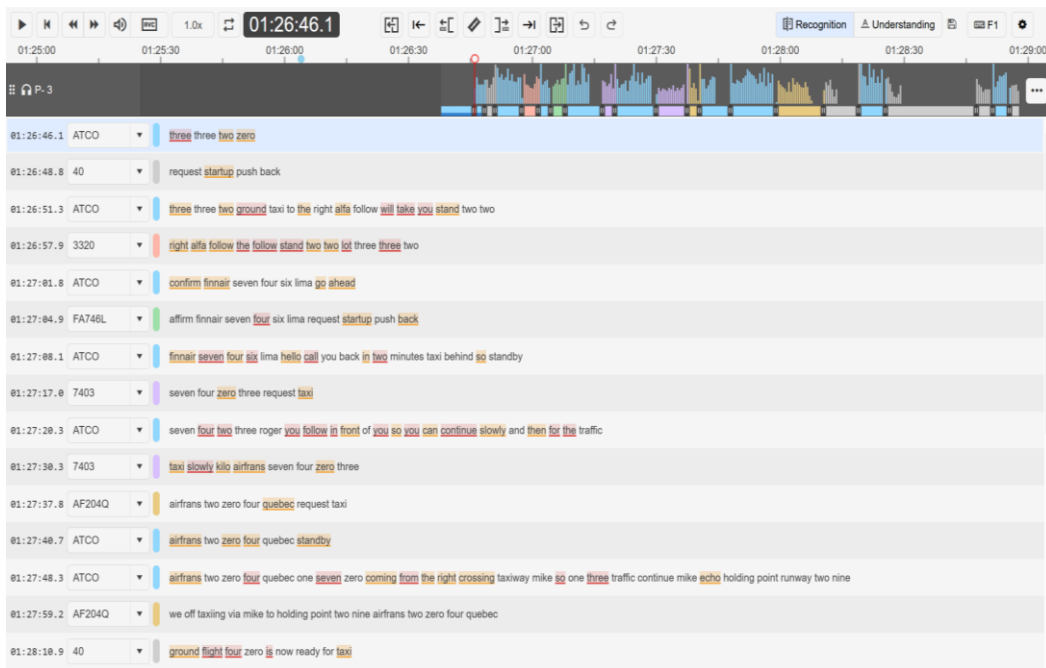
Customer environment changes over time

- Customer environment will change over the lifetime:
- New ATCO's, radios, microphones...
- Waypoints are changed on a regular basis



For increased automation levels a WER <10% is needed

What exactly did we build? Offline ASR for Operations



- Transcription of specific channels & timeframes
- Multi-speaker segmentation
- Human correction & verification workflow
- Export to incident reports

How do we evaluate ASR systems?

↓ Word Error Rate (WER):



predicted: lufthansa two zero one there's traffic from the right

true: lufthansa two zero one there's traffic from the left

WER = 0.1

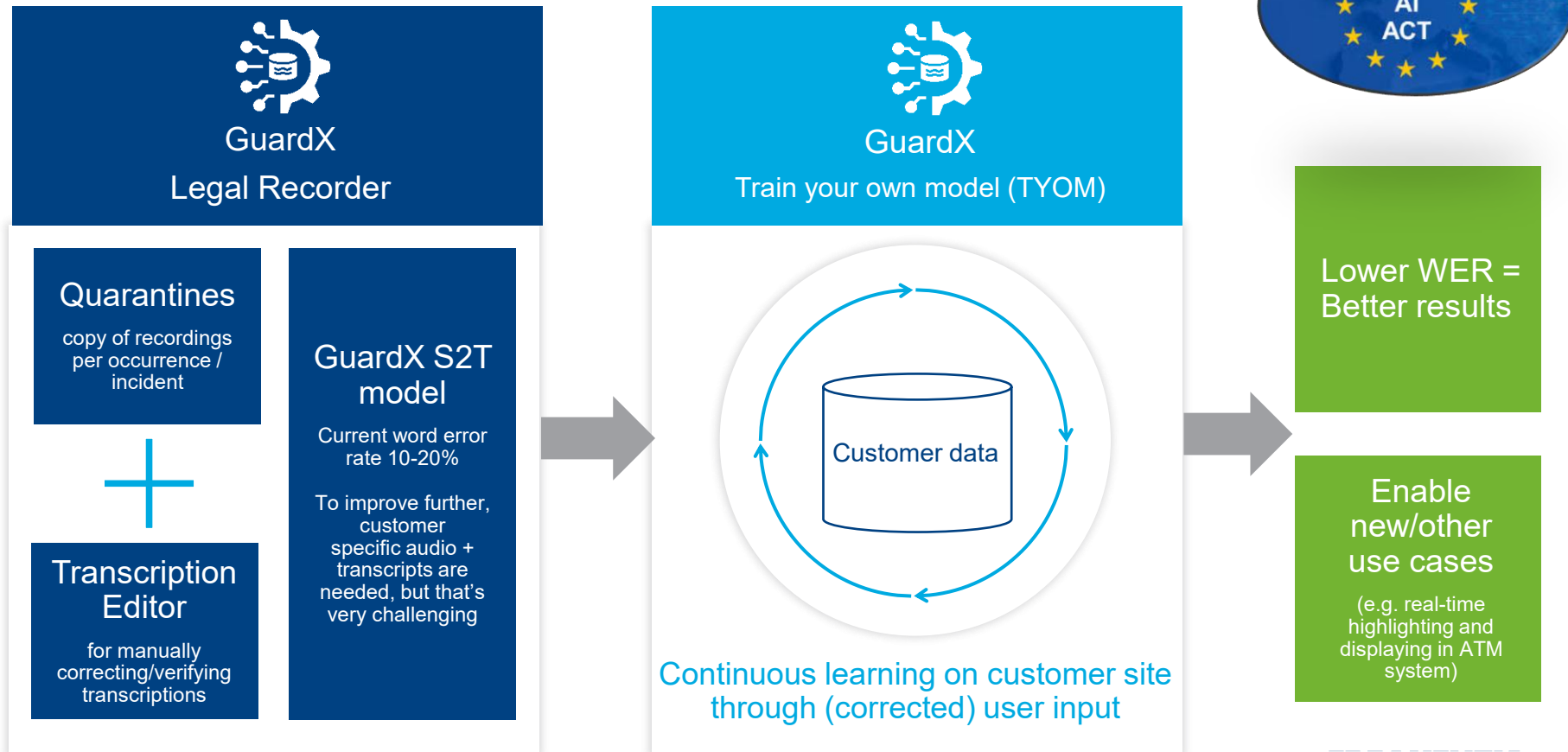
↑ Inverse Real Time Factor (RTFx)

↑ Entity correctness: callsigns, waypoints, flight levels



https://huggingface.co/spaces/hf-audio/open_asr_leaderboard

What exactly did we build? Train Your Own Model



GuardX TYOM (Data selection)

- Labeled data = **audio segment + corrected transcript** produced during investigation work
- **No additional labeling cost**
- **Minimum ~10 minutes (better ≥ 30 minutes)**
- **Key constraint: customer controls data** (*privacy/security*)

The screenshot displays the GuardX Administration interface. The top navigation bar includes 'Activity', 'Quarantines', 'Monitoring', and 'Administration'. The left sidebar contains various system management options, with 'AI Training' highlighted. The main content area is titled 'Training Sets' and lists several training sets, including 'TYOM Demo Set' and 'NewATCOs'. The 'TYOM Demo Set' is selected, showing its details on the right. The details include a 'Name' field with the value 'TYOM Demo Set', a 'Base Model' dropdown menu, an 'Only Verified Transcription' checkbox, and a 'Timespan' field set to '2025-01-01' to '2025-02-28'. A 'Save' button and a link to 'Show Advanced Configuration' are also visible.

Training Sets
TYOM Demo Set
17 Mar 2026 07:46:16
09 Mar 2026 14:02:33
09 Mar 2026 13:15:58
08 Mar 2026 20:38:49
25 Feb 2026 14:39:04
24 Feb 2026 11:13:52
18 Feb 2026 13:11:51
17 Feb 2026 08:19:42
16 Feb 2026 10:51:36
Training Set 2
06 Nov 2025 14:14:19
NewATCOs

Name:
TYOM Demo Set
A name for the training set. This name will be used in the model name for models trained from this dataset.

Base Model:
[Dropdown menu]
The model to use as a starting point for the training process. This must be a base model and cannot be a model that has already been trained.

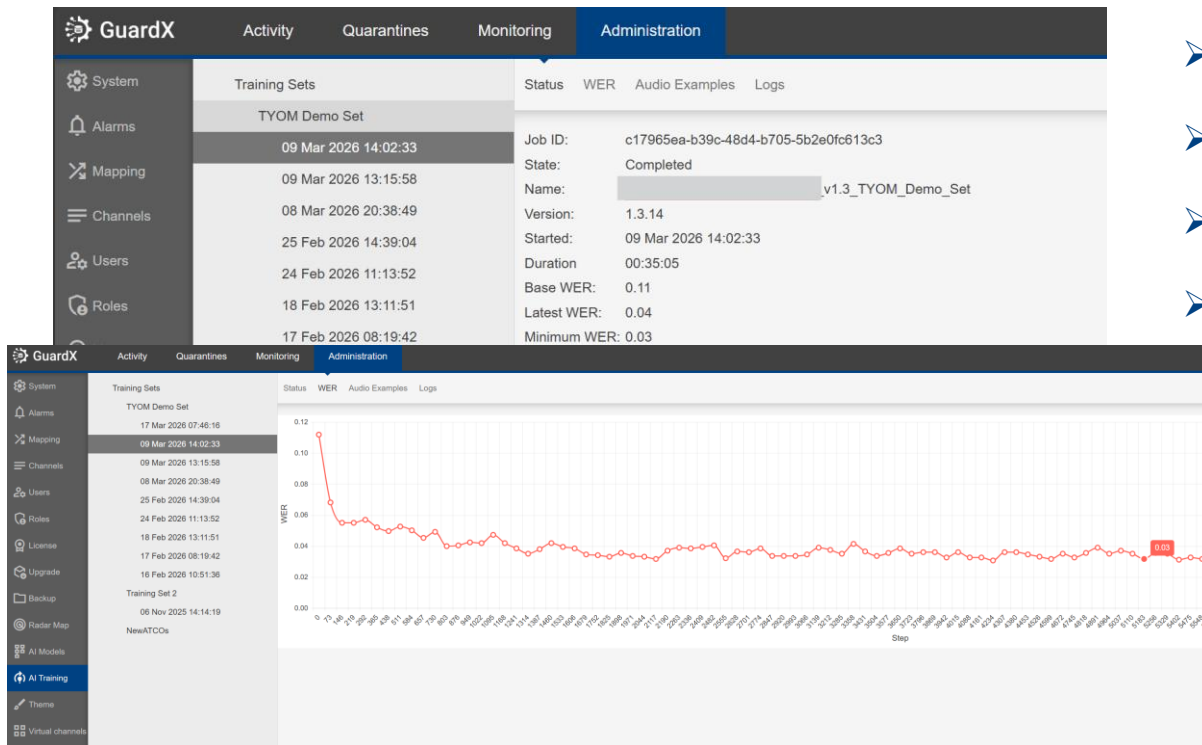
Only Verified Transcription: ☐
If this is checked, only transcriptions that are verified as correct will be included for training. Otherwise all transcriptions will be used.

Timespan:
2025-01-01 - 2025-02-28
Limits the data used for training. Transcriptions from quarantines with a start time that falls within the selected period will be used for training. The end date can be left empty to use all quarantines after the start date. The period starts at midnight UTC inclusive on the start date, and ends just before midnight UTC on the date following the end date.

[Save](#)

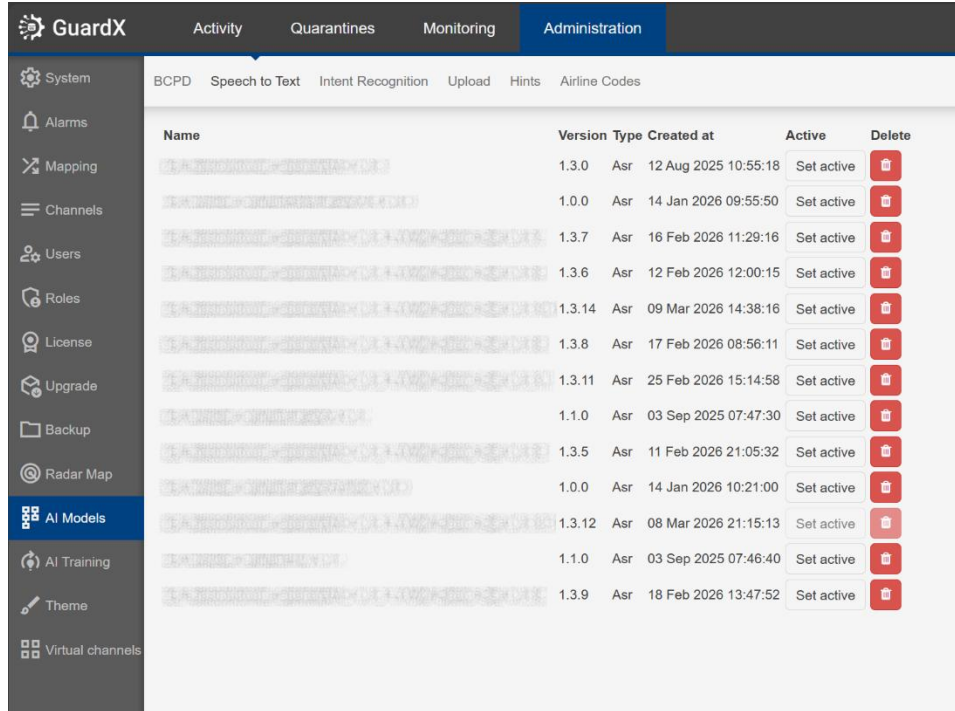
[Show Advanced Configuration](#)

GuardX TYOM (Training execution)



- **Container** based trainings
 - Runs on **auxiliary infrastructure**
 - GPU: **~20–30m** | CPU: **~20–30h**
 - Training progress and improvement can be monitored
- Model remains **locally** and is customer-controlled

GuardX TYOM (Model selection)



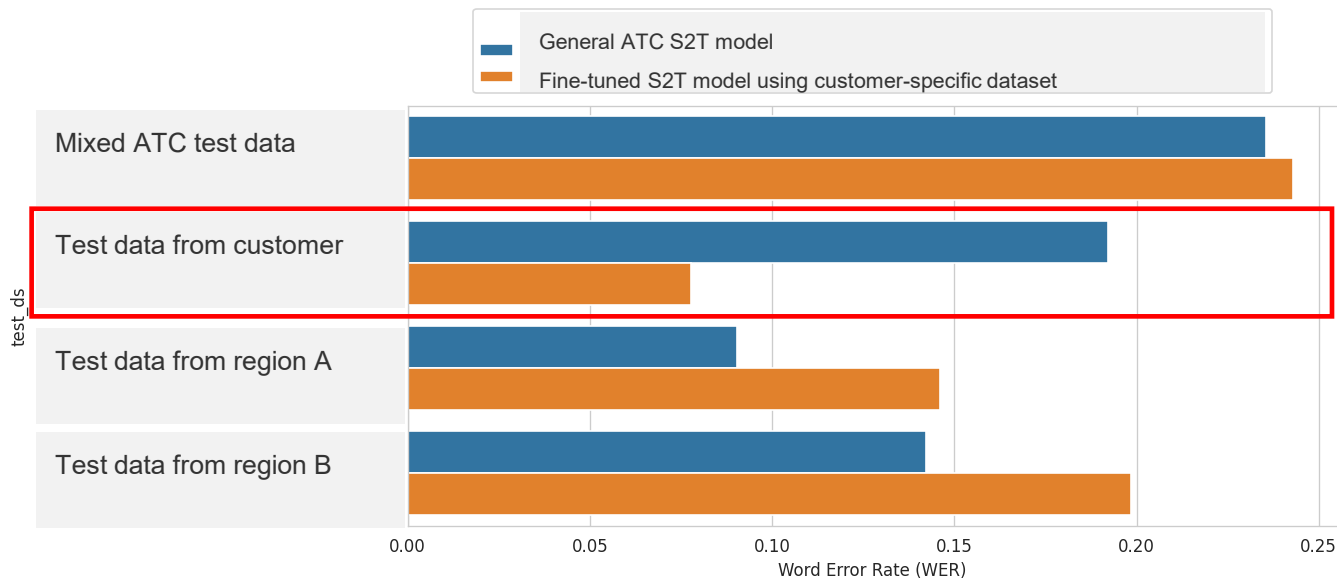
The screenshot displays the GuardX Administration interface. The left sidebar contains navigation links for System, Alarms, Mapping, Channels, Users, Roles, License, Upgrade, Backup, Radar Map, AI Models (highlighted), AI Training, Theme, and Virtual channels. The main content area is titled 'Administration' and shows a table of AI Models. The table has columns for Name, Version, Type, Created at, Active, and Delete. The 'AI Models' section is active, showing a list of models with their respective versions and creation dates. The 'Active' column shows 'Set active' for each model, and the 'Delete' column shows a trash icon.

Name	Version	Type	Created at	Active	Delete
1.3.0	Asr	12 Aug 2025 10:55:18	Set active		
1.0.0	Asr	14 Jan 2026 09:55:50	Set active		
1.3.7	Asr	16 Feb 2026 11:29:16	Set active		
1.3.6	Asr	12 Feb 2026 12:00:15	Set active		
1.3.14	Asr	09 Mar 2026 14:38:16	Set active		
1.3.8	Asr	17 Feb 2026 08:56:11	Set active		
1.3.11	Asr	25 Feb 2026 15:14:58	Set active		
1.1.0	Asr	03 Sep 2025 07:47:30	Set active		
1.3.5	Asr	11 Feb 2026 21:05:32	Set active		
1.0.0	Asr	14 Jan 2026 10:21:00	Set active		
1.3.12	Asr	08 Mar 2026 21:15:13	Set active		
1.1.0	Asr	03 Sep 2025 07:46:40	Set active		
1.3.9	Asr	18 Feb 2026 13:47:52	Set active		

- Deploy trained model into GuardX
- Switch between models
- **Validate** before **activation**
- Supports safe usage aligned with operational workflows

Why domain adaption matters

- Largest accuracy gains appear on **customer-specific test data**



Why it's still hard: failure modes

Failure Mode	Cause	Mitigation
 Hallucination	 Unconstrained decoding / noise	 Using architecture robust against hallucination

text sydney tower good morning jetstar five hundred

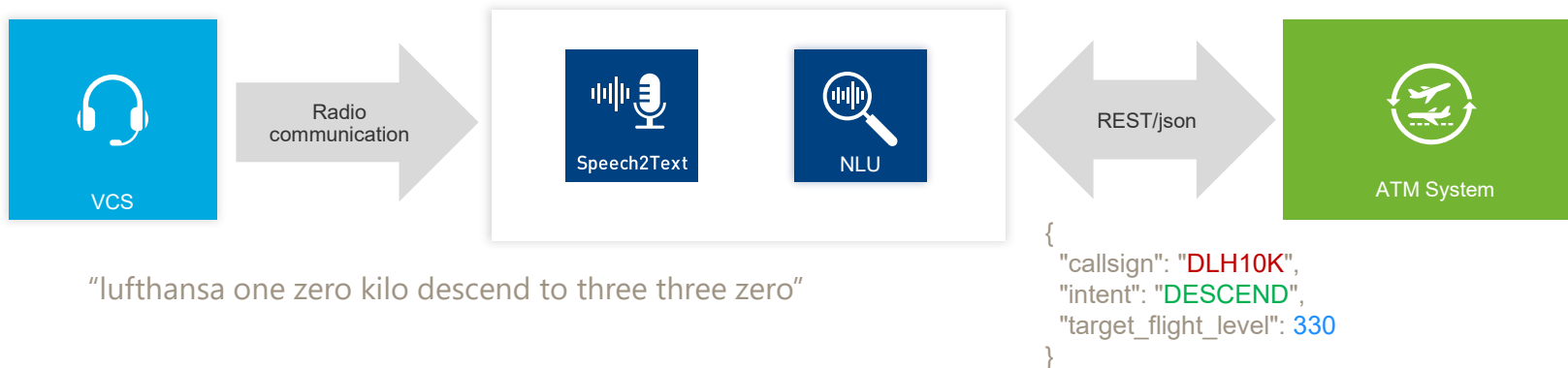
pred text tun tower good morning four four four four four four four four four four four four four four four four four four
four
four four four four four four four four four four four four four four four four four four four two tower good morning
just for the tower good morning just for the tower good morning just for one one one one one one one one one one one
one one

WER 2014.29

 Multilingual Mixing	 Code-switching / Accents	models supporting multilingual languages
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Future roadmap: From transcription → understanding

GuardX Voice2Intention – Real time automatic speech recognition for ATM



Aligned with EUROCAE WG-126, which Aims for standardised ATC and VCS integration to enhance controller awareness, reduce workload, and improve flight safety



EASA classification: Human-AI Teaming (HAT)

Safety Impact: Medium - High (depending on the use case)

Failure Modes: Several (depending on the use case)

Key takeaways

Generic ASR fails
in safety-critical
domains

Data > model

Human-in-the-
loop is non-
negotiable

MLOps is the real
system challenge

Let's discuss 🚀

